Experiment 20: Person DOB Knowledge Base

Aim:

To implement a knowledge base of persons with date of birth and answer queries using logical reasoning.

Algorithm:

1. Store person facts with DOB.
2. Support queries by name or by birth year/month.
3. Compute age or filter persons based on criteria.

Code:

```
from datetime import datetime

people = {
    'Alice': '1995-03-15',
    'Bob': '1988-11-22',
    'Charlie': '2001-07-08',
    'Diana': '1995-06-30'
}

def born_in_year(year):
    return [name for name, dob in people.items() if dob.startswith(str(year))]

def age(name):
    dob = datetime.strptime(people[name], '%Y-%m-%d')
    today = datetime(2026, 10, 16)
    return today.year - dob.year - ((today.month, today.day) < (dob.month, dob.day))

print("Born in 1995:", born_in_year(1995))
print("Alice's age:", age('Alice'))
```

Sample Output:

```
Born in 1995: ['Alice', 'Diana']
Alice's age: 31
```

Result:

The DOB knowledge base correctly answered queries about birth year and calculated ages from stored facts.